

Fathom common bean

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Abstract: Fathom (CFIA registration no. 7544) is a full-season navy bean (*Phaseolus vulgaris* L.) cultivar with a high yield potential, and resistance to anthracnose (caused by *Colletotrichum lindemuthianum*) and common bacterial blight (CBB; caused by *Xanthomonas axonopodis* pv. *phaseoli*). Fathom is recommended for the dry bean growing areas in southwestern Ontario.

Key words: *Phaseolus vulgaris* L., navy bean, common bacterial blight resistance, anthracnose.

Résumé : Fathom (numéro d'homologation 7544 de l'ACIA) est une variété tardive de haricot (*Phaseolus vulgaris* L.) se caractérisant par un rendement potentiel élevé et la résistance à l'anthracnose (causée par *Colletotrichum lindemuthianum*) ainsi qu'à la brûlure bactérienne (causée par *Xanthomonas axonopodis* pv. *phaseoli*). On recommande Fathom pour les zones du sud-ouest de l'Ontario servant à la production de haricot sec. [Traduit par la Rédaction]

Mots-clés : *Phaseolus vulgaris* L., haricot, résistance à la brûlure bactérienne, anthracnose.

Introduction

Fathom is an upright navy bean (*Phaseolus vulgaris* L.) cultivar with an excellent yield, full-season maturity, and resistance to anthracnose (caused by race 73 of *Colletotrichum lindemuthianum*) and common bacterial blight (CBB; caused by *Xanthomonas axonopodis* pv. *phaseoli*). Fathom was developed by the University of Guelph, Guelph, ON, Canada and was tested in the Ontario White Bean Registration and Performance Trials in 2012, 2013, and 2014. Fathom was registered by the Variety Registration Office, Canadian Food Inspection Agency, Ottawa, ON, on 7 May 2014 (Registration number 7544).

Parentage and Ancestry

Fathom was developed from the cross of spse087 and spsbf019, which are sister lines selected in F₆ generation as single plants from the bi-parental cross of OAC Rex/OAC Seaforth. OAC Rex was used to provide upright architecture with high podding and narrow canopy (Michaels et al. 2006). OAC Seaforth originated from a cross between Seafarer and PI 326418, backcrossed to Seafarer. PI 326418 was used as a source of the Co-2 (the *Are* gene), which confers resistance to several races of *C. lindemuthianum* (Beversdorf and Buzzell 1984). The

sister line spse087 was selected for its early maturity and another sister line spsbf019 was selected due to its resistance to common bacterial blight.

Breeding Method

The breeding population from the sister line cross was advanced one generation per year using a modified-bulk method, in which single pods were harvested from single plants in each generation and bulked to plant the next generation, to the F₅ generation at the Elora Research Station, ON. The F₅ bulk population was planted in the summer of 2007 at the Elora Research Station, ON. Single plants were selected and F_{5.6} plant-rows were grown in 2008 at Elora Research Station. Criteria for selection of single plants and plant-rows were maturity, absence of anthracnose and common bacterial blight symptoms, high yield potential, and upright plant type. Seeds of the selected single plants and plant-rows were also examined for seed size, shape, and colour to conform to the standards of the navy bean market class. The lines were further tested and selected in the replicated preliminary yield trials at the Elora Research Station in 2009. The lines were entered into advanced yield trials at the Elora Research Station and a field location near St. Thomas, ON, in 2010.

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Performance

Fathom was entered into the Ontario White Bean Registration and Performance Trials as ACUG 12-5 and was evaluated in multi-location yield-trials across Ontario in 2012, 2013, and 2014. These tests were performed under the guidelines set by the Ontario Pulse Committee (www.gobeans.ca) with four replications per location. Test locations with coefficients of variation (CVs) lower than 15% were considered valid tests. Yield (adjusted to 18% moisture after combine harvest), days to maturity (number of days from planting to maturity), and 100-seed weight were collected on a plot basis. The data were subjected to an analysis of variance to determine the least square means and least significant difference (LSD) for each genotype within and among locations. Cooking quality tests were performed with a composite seed samples from three locations by mixing 200 g of seed of each replicate for each entry. These samples were processed in the Food Pilot Plant at the Agriculture and Agri-Food Canada, Lethbridge Research Center, Alberta and evaluated for cooking and canning quality parameters such as hydration coefficient, washed-drained weight, and texture of the canned beans. The texture of the canned beans was evaluated using wire extrusion cells by the Instron texture measurement system Model 441 (Instron Corporation, Canton, MA) for firmness and plateau force (Voisey 1971). A detail canning quality evaluation method is described in Navabi et al. (2013).

Common bacterial blight severity ratings were taken in an artificially inoculated disease nursery in AAFC, Harrow, ON. The lines were inoculated at the fourth trifoliate leaf stage with spore suspensions of three isolates of *X. axonopodis* pv. *Phaseoli* (18, 118, and 98) by spraying the leaves with a suspension of 107 colony-forming units (CFU) mL⁻¹ inoculum. The severity of CBB was visually estimated as percentage of necrotic leaf area in the plot, rated 14 and 21 days after inoculation (Mutlu et al. 2005). The disease severity was rated based on leaf area infection with a 0 to 5 scale where 0 = no symptoms, 1 = <5%, 2 = 5–10%, 3 = 10–25%, 4 = 25–50%, and 5 = 50–100% (Yu et al. 2000). Anthracnose caused by [*Colletotrichum lindemuthianum* (Sacc. & Magnus)] and bean common mosaic virus (BCMV) symptoms were evaluated after inoculation of 10–15 plants grown in 1 L pots filled with ProMix at the AAFC, Harrow, ON. Anthracnose inoculation was done according to the method of Balardin et al. (1997) by brushing both the upper and lower surfaces of fully expanded primary leaves of 7–10 day-old seedlings with *C. lindemuthianum* spores (10⁶ spores mL⁻¹) cultured in Mathur's medium. Inoculated plants were placed in a mist chamber, 100% humidity at 23°C for 48 h, and then transferred to a growth cabinet at 23/18°C day/night temperature with a 14 h photoperiod. Disease rating was done five days after inoculation using a visual score of 1–9, with 9 being the

most susceptible and was repeated three days later (Corrales and van Schoonhoven 1987). Inoculation for BCMV was done on 10-day-old plants or when unifoliate leaves were fully expanded. The inoculum was maintained by multiple cycles of harvesting seeds from infected plants. For inoculation, three to five infected leaves were ground using a mortar and pestle in a 0.01 molar phosphate buffer (pH = 7.0) with an addition of carborundum powder (Fisher Scientific, Pittsburgh, PA). The upper surfaces of the leaves were covered with the ground infected leaves. Once the surfaces were dry the plants were misted with water and maintained at 23/18°C day/night temperature with a 14 h photoperiod. Beginning 10 days after inoculation, seeding showing clear signs of mosaic or systemic necrosis were counted and removed and scored as susceptible. Beginning 20 days after inoculation, plants clearly not affected by necrosis are counted and removed and scored as resistant (Johnson et al. 1997).

Agronomic Performance and Yield

Across 15 locations in the Ontario White Bean Registration and Performance Trials during 2012, 2013, and 2014, Fathom navy bean yielded 3675 kg ha⁻¹, which was 226 kg higher than the check cultivar mean (AC Compass, OAC Thunder, Lightning, and Rexeter) (Table 1). Fathom was rated as a full-season maturity cultivar with days to maturity significantly earlier (three days) than check cultivar Rexeter but one day later than other check mean (AC Compass, OAC Thunder and Lightning). It had a 100-seed weight larger than all checks in 2012 and 2013 (Table 1).

In canning evaluations, both hydration coefficient and degree of packing were similar to check cultivars, but Fathom had a slightly lower firmness than checks in both years (Table 2). The washed drained weight was higher for Fathom than the checks in 2012 but similar to the checks in 2013.

Based on growth room and field observations, Fathom is resistant to anthracnose race 73 and BCMV race 1 and race 15 (Table 3). Fathom possesses the SCAR marker SW13 (Melotto et al. 1996) linked to resistance to race 1 and 15 of BCMV. It has moderate levels of resistance to CBB which is similar to Rexeter (Smith et al. 2012). Fathom carries the CBB resistance SCAR marker SU91 for the major CBB resistance QTL (Pedraza et al. 1997) on chromosome Pv08.

Fathom has an indeterminate growth habit with erect branching and an upright plant type with a long vine and high podding nodes and good lodging resistance. It has green hypocotyls and white flowers. Plants have dark green leaf colour. The pods are light tan coloured when ripe and covered with short pubescence. Seeds are white with dull seed coat lustre and white hilum. Fathom is adapted to and recommended for bean

Table 1. Yield, days to maturity, and seed weight of Fathom navy bean compared with check cultivars, tested in the Ontario White Bean Registration and Performance Trials from 2012 to 2014^a.

Cultivars	Yield (kg ha ⁻¹)				Days to maturity				100 Seed weight (g)			
	2012	2013	2014	Mean	2012	2013	2014	Mean	2012	2013	2014	Mean
AC Compass	2968	3544	3263	3258	93.3	102.6	104.1	100	22.8	19.6	21.9	21.4
OAC Thunder	3368	3612	3265	3415	95.5	102.4	104.9	100.9	23.9	23	21.61	22.9
Lightning	3186	3575	3417	3393	93.4	102.4	105.9	100.6	23.7	21.8	23.58	23
Rexeter	3376	3959	3860	3732	100.8	107.6	110	106.1	22.2	21.1	23.72	22.3
Fathom	3625	3944	3456	3675	99	104.9	105.3	103.1	24.6	23.3	20.89	22.9
No. of trials	6	4	5		6	4	5		5	4	4	
LSD 5%	174	178	299		1.6	0.8	0.5		0.6	0.5	1.17	

^aTest locations were Blyth, Elora, Kippen, St. Thomas, Woodstock, and Highbury in 2012, Blyth, Elora, Kippen, and St. Thomas in 2013, and Auburn, Elora, Highbury, St. Thomas, and Woodstock in 2014.

Table 2. Cooking quality of canned beans of Fathom compared with check cultivars grown in the Ontario White Bean Registration and Performance Trials in 2012 and 2013, averaged over three locations.

Cultivars	Hydration coefficient ^a		Degree of packing (1–5) ^b		Washed drained wt. (%) ^c		Texture measurement ^d			
							Plateau force (N)		Firmness (N mm ⁻¹)	
	2012	2013	2012	2013	2012	2013	2012	2013	2012	2013
AC Compass	2.2	2.3	1.0	1.3	59.88	60.33	318	479	20.49	25.33
Lightning	2.2	2.3	1.3	1.7	58.64	59.00	290	435	19.10	25.33
OAC Thunder	2.2	2.2	1.0	1.3	57.36	58.33	321	491	22.13	26.76
Nautica	2.3	2.3	1.3	1.3	58.64	59.00	277	497	19.84	29.01
Fathom	2.2	2.2	1.0	1.7	61.46	59.33	311	533	13.14	22.33
SE	0.02	0.07	0.50	0.78	1.90	1.49	60	76	7.41	2.92

^aSoaked wt. (blanched in 88°C water for 45 min) divided by dry weight (determined for 500 g of beans).

^bScored visually; 1: no clumping and 5: over half clumped.

^cWeight of beans after washed and drained on a screen, presented as percentage of unwashed-undrained weight.

^dTexture of canned beans was measured on Instron Texture measurement system using wire extrusion cells.

Table 3. Response of Fathom to anthracnose, bean common mosaic virus (BCMV), and common bacterial blight (CBB) compared with check cultivars.

Cultivar	Anthracnose ^a	BCMV ^b		CBB (0–5) ^c	
	Race 73	Race 1	Race 15	2012	2013
Rexeter	S	R	S	2.3	2.5
Lighthouse	S	R	R	1.8	2.5
Fathom	R	R	R	3.5	3

^aReactions against Anthracnose race 73 were assessed after artificial inoculation under controlled conditions using a visual score of 1–9, with 9 being the most susceptible (Corrales and Schoonhoven 1987).

^bReactions against bean common mosaic virus race 1 and 15 were assessed after artificial inoculation under controlled conditions as susceptible (S) or resistant (R).

^cCommon bacterial blight severity was rated based on leaf area infection with a 0 to 5 scale where 0 = no symptoms, 1 = <5%, 2 = 5–10%, 3 = 10–25%, 4 = 25–50%, and 5 = 50–100% (Yu et al. 2000).

growing areas in Ontario with 2700 or more crop heat unit areas.

Maintenance and Distribution of Pedigreed Seed

In 2012, Fathom was planted in isolation plots for purification and multiplication of seed and then in a disease-free environment in Idaho in 2013 for pre-breeder and breeder seed production. University of Guelph, Bean breeding program, Guelph, ON N1G 2W1, Canada, will maintain the breeder seed. Pedigreed seed will be distributed by Hensall District Co-operative (HDC), 1 Davidson Drive, P.O. Box 219, Hensall, ON N0M 1X0, Canada, Phone: 519-262-3002, Fax: 519-262-2317.

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References

- Balardin, R.S., Jarosz, A.M., and Kelly, J.D. 1997. Virulence and molecular diversity in *Colletotrichum lindemuthianum* from South, Central and North America. *Phytopathology*, **87**: 1184–1191. doi:[10.1094/PHYTO.1997.87.12.1184](https://doi.org/10.1094/PHYTO.1997.87.12.1184).
- Berversdorf, W.D., and Buzzell, R.I. 1984. OAC Seaforth field bean. *Can. J. Plant Sci.* **64**: 757–758. doi:[10.4141/cjps84-103](https://doi.org/10.4141/cjps84-103).
- Corrales, M.A.P., and van Schoonhoven, A. 1987. Standard system for the evaluation of bean germplasm. Centro Internacional de Agricultura Tropical (CIAT), Cali, Colombia.
- Johnson, W.C., Guzmán, P., Mandala, D., Mkandawire, A.B.C., Temple, S., Gilbertson, R.L., and Gepts, P. 1997. Molecular tagging of the bc-3 gene for introgression into Andean common bean. *Crop Sci.* **37**: 248–254. doi:[10.2135/cropsci1997.0011183X003700010044x](https://doi.org/10.2135/cropsci1997.0011183X003700010044x).
- Melotto, M., Afanador, L., and Kelly, J.D. 1996. Development of a SCAR marker linked to the *I* gene in common bean. *Genome*, **39**: 1216–1219. doi:[10.1139/g96-155](https://doi.org/10.1139/g96-155).
- Michaels, T.E., Smith, T.H., Larsen, J., Beattie, A.D., and Pauls, K.P. 2006. OAC Rex common bean. *Can. J. Plant Sci.* **86**: 733–736. doi:[10.4141/P05-128](https://doi.org/10.4141/P05-128).
- Mutlu, N., Miklas, P., Reiser, J., and Coyne, D. 2005. Backcross breeding for improved resistance to common bacterial blight in pinto bean (*Phaseolus vulgaris* L.). *Plant Breed.* **124**: 282–287. doi:[10.1111/j.1439-0523.2005.01078.x](https://doi.org/10.1111/j.1439-0523.2005.01078.x).
- Navabi, A., Rupert, T., Park, S.J., Yu, K., Smith, T.H., and Pauls, K.P. 2013. Apex common bean. *Can. J. Plant Sci.* **93**: 131–135. doi:[10.4141/cjps2012-139](https://doi.org/10.4141/cjps2012-139).
- Pedraza, F.G., Gallego, G., Beebe, S., and Tohme, J. 1997. Marcadores SCAR y RAPD para la resistencia a la bacteriosis comun (CBB). Pages 130–134 in S.P. Singh and O. Voysest, eds. Taller de mejoramiento de frijol para el Siglo XXI: Bases para una estrategia para America Latina. Centro Internacional de Agricultura Tropical (CIAT), Cali, Colombia.
- Smith, T.H., Michaels, T.E., Navabi, A., and Pauls, K.P. 2012. Rexeter common bean. *Can. J. Plant Sci.* **92**: 351–353. doi:[10.4141/cjps2011-184](https://doi.org/10.4141/cjps2011-184).
- Voisey, P.W. 1971. The Ottawa texture measuring system. *J. Can. Inst. Food. Technol.* **4**: 91–103. doi:[10.1016/S0008-3860\(71\)74189-0](https://doi.org/10.1016/S0008-3860(71)74189-0).
- Yu, K., Park, S.J., and Poysa, V. 2000. Marker-assisted selection of common beans for resistance to common bacterial blight: efficacy and economics. *Plant Breed.* **119**: 411–415. doi:[10.1046/j.1439-0523.2000.00514.x](https://doi.org/10.1046/j.1439-0523.2000.00514.x).